# Configuring PPP to Connect the Headquarters and Branch

Difficulty (out of 3 stars): ★★★ Recommended study time: 1 hour

## Contents

# 1 About This Lab

## 1.1 Task Description

The Point-to-Point Protocol (PPP) is a commonly used WAN data link layer protocol that provides point-to-point data encapsulation and transmission over full-duplex links. PPP supports Password Authentication Protocol (PAP) authentication and Challenge Handshake Authentication Protocol (CHAP) authentication.

After completing this task, you will gain a clear understanding of PPP and learn how to configure PPP and PPP authentication.

## 1.2 Objectives

On completing this task, you will be able to:

1. Master the basic PPP configuration.
2. Configure PAP authentication for a PPP link.
3. Configure CHAP authentication for a PPP link.

# 2 Task Preparation

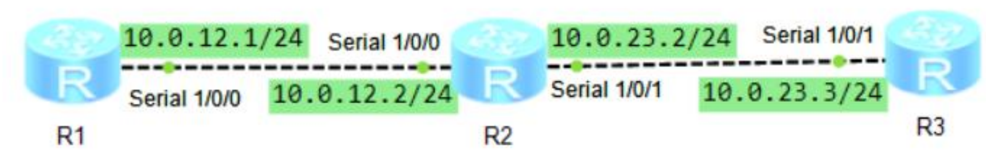
## 2.1 Preparatory Knowledge

You need to learn basic network knowledge before starting this lab. You can learn this course through the following materials:

1. Training material: HCIA-Datacom Network Technology Learning Guide
2. Online course: HCIA-Datacom V1.0 Huawei Certified Datacom Engineer Online Course

## 2.2 Network Topology

**Figure 2-1** Lab topology for configuring PPP to connect the headquarters and branch




R2 is deployed at the headquarters of a company. R1 and R3 are deployed at two branches of the company. You need to connect the headquarters network and branch networks over a WAN. Use HDLC and PPP on WAN links and configure different authentication modes to ensure security.

## 2.3 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
```

```
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int s1/0/0
[R1-Serial1/0/0]ip address 10.0.12.1 24
[R1-Serial1/0/0]quit
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int s1/0/0
[R2-Serial1/0/0]ip address 10.0.12.2 24
[R2-Serial1/0/0]quit
[R2]int s1/0/1
[R2-Serial1/0/1]ip address 10.0.23.2 24
[R2-Serial1/0/1]quit
[R2]
```

- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]int s1/0/1
[R3-Serial1/0/1]ip address 10.0.23.3 255.255.255.0
[R3-Serial1/0/1]quit
[R3]
```

# 3 Task Implementation

## 3.1 Checking Basic Configurations

Omitted

## 3.2 Setting PPP as the Encapsulation Type of a Serial Interface

The default encapsulation type of serial interfaces on Huawei routers is PPP, so no configuration is required. Check the encapsulation type.

```
<R1>display current-configuration int s1/0/0
[V200R003C00]
#
interface Serial1/0/0
 link-protocol ppp
 ip address 10.0.12.1 255.255.255.0
#
return
<R1>

<R2>display current-configuration int s1/0/0
```

```
[V200R003C00]
#
interface Serial1/0/0
 link-protocol ppp
 ip address 10.0.12.2 255.255.255.0
#
return
<R2>display current-configuration int s1/0/1
[V200R003C00]
#
interface Serial1/0/1
 link-protocol ppp
 ip address 10.0.23.2 255.255.255.0
#
return
<R2>

<R3>display current-configuration int s1/0/1
[V200R003C00]
#
interface Serial1/0/1
 link-protocol ppp
 ip address 10.0.23.3 255.255.255.0
#
return
<R3>
```

## 3.3 Checking the Link Connectivity After Completing the Configuration

```
<R2>ping -c 2 10.0.12.1
  PING 10.0.12.1: 56  data bytes, press CTRL_C to break
    Reply from 10.0.12.1: bytes=56 Sequence=1 ttl=255 time=40 ms
    Reply from 10.0.12.1: bytes=56 Sequence=2 ttl=255 time=30 ms

  --- 10.0.12.1 ping statistics ---
    2 packet(s) transmitted
    2 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 30/35/40 ms

<R2>ping -c 2 10.0.23.3
  PING 10.0.23.3: 56  data bytes, press CTRL_C to break
    Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=30 ms
    Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=255 time=40 ms

  --- 10.0.23.3 ping statistics ---
    2 packet(s) transmitted
    2 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 30/35/40 ms
```

```
<R2>
```

## 3.4 Configuring OSPF

1. Enable OSPF on all the three routers and configure them to advertise their direct routes.

```
[R1]ospf 1
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0]network   10.0.12.0 0.0.0.255
[R1-ospf-1-area-0.0.0.0]quit
[R1-ospf-1]quit
[R1]

[R2]ospf 1
[R2-ospf-1]area 0
[R2-ospf-1-area-0.0.0.0]network   10.0.12.0 0.0.0.255
[R2-ospf-1-area-0.0.0.0]network   10.0.23.0 0.0.0.255
[R2-ospf-1-area-0.0.0.0]quit
[R2-ospf-1]quit
[R2]

[R3]ospf 1
[R3-ospf-1]area 0
[R3-ospf-1-area-0.0.0.0]network   10.0.23.0 0.0.0.255
[R3-ospf-1-area-0.0.0.0]
[R3-ospf-1-area-0.0.0.0]quit
[R3-ospf-1]quit
[R3]
```

2. After completing the configuration, check whether the devices have learned the corresponding routes through OSPF.

```
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 9          Routes : 9

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      10.0.12.0/24  Direct  0    0           D   10.0.12.1       Serial1/0/0
      10.0.12.1/32  Direct  0    0           D   127.0.0.1       Serial1/0/0
      10.0.12.2/32  Direct  0    0           D   10.0.12.2       Serial1/0/0
    10.0.12.255/32  Direct  0    0           D   127.0.0.1       Serial1/0/0
      10.0.23.0/24  OSPF    10   96          D   10.0.12.2       Serial1/0/0
        127.0.0.0/8 Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[R1]
```

3. Verify that the corresponding routes have been learned through OSPF.

On R1, run the **ping** command to check the connectivity between R1 and R3.

```
[R1]ping -c 2 10.0.23.3
  PING 10.0.23.3: 56   data bytes, press CTRL_C to break
    Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=254 time=40 ms
    Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=254 time=30 ms

  --- 10.0.23.3 ping statistics ---
    2 packet(s) transmitted
    2 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 30/35/40 ms

[R1]
```

## 3.5 Enabling PAP Authentication on R1 and R2

1. Configure PAP authentication and configure R1 as the PAP authenticator.

```
[R1]interface Serial 1/0/0
[R1-Serial1/0/0]
[R1-Serial1/0/0]ppp authentication-mode pap
[R1-Serial1/0/0]
[R1-Serial1/0/0]quit
[R1]
[R1]aaa
[R1-aaa]
[R1-aaa]local-user huawei password cipher huawei123
Info: Add a new user.
[R1-aaa]
[R1-aaa]local-user huawei service-type ppp
[R1-aaa]
[R1-aaa]
[R1-aaa]quit
[R1]
```

2. Configure R2 as the PAP supplicant.

```
[R2]interface Serial 1/0/0
[R2-Serial1/0/0]
[R2-Serial1/0/0]ppp pap local-user huawei password cipher huawei123
[R2-Serial1/0/0]
[R2-Serial1/0/0]quit
[R2]
```

3. After completing the configuration, check the connectivity between R1 and R2.

```
[R1]ping 10.0.12.2
  PING 10.0.12.2: 56   data bytes, press CTRL_C to break
    Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=30 ms
    Reply from 10.0.12.2: bytes=56 Sequence=2 ttl=255 time=30 ms
    Reply from 10.0.12.2: bytes=56 Sequence=3 ttl=255 time=20 ms
    Reply from 10.0.12.2: bytes=56 Sequence=4 ttl=255 time=40 ms
    Reply from 10.0.12.2: bytes=56 Sequence=5 ttl=255 time=20 ms

  --- 10.0.12.2 ping statistics ---
```

```
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/28/40 ms

[R1]
```

## 3.6 Enabling CHAP Authentication on R2 and R3

1. Configure R3 as the CHAP authenticator.

```
[R3]interface Serial 1/0/1
[R3-Serial1/0/1]ppp authentication-mode chap
[R3-Serial1/0/1]quit
[R3]
[R3]aaa
[R3-aaa]local-user huawei password cipher huawei123
Info: Add a new user.
[R3-aaa]local-user huawei service-type ppp
[R3-aaa]quit
[R3]
[R3]interface Serial 1/0/1
[R3-Serial1/0/1]shutdown
[R3-Serial1/0/1]undo shutdown
[R3-Serial1/0/1]quit
[R3]
```

Note that the following information is displayed on R3:

```
[R3]
Apr   5 2025 19:27:55-08:00 R3 %%01PPP/4/PEERNOCHAP(l)[7]:On the interface Serial
1/0/1, authentication failed and PPP link was closed because CHAP was disabled o
n the peer.
[R3]
Apr   5 2025 19:27:55-08:00 R3 %%01PPP/4/RESULTERR(l)[8]:On the interface Serial1
/0/1, LCP negotiation failed because the result cannot be accepted.
[R3]
```

In the command output, the information in bold indicates that the authentication fails.

2. Configure R2 as the CHAP supplicant.

```
[R2]interface Serial 1/0/1
[R2-Serial1/0/1]ppp chap user huawei
[R2-Serial1/0/1]ppp chap password cipher huawei123
[R2-Serial1/0/1]quit
[R2]
```

After the configuration is complete, the interface becomes up. Run the **ping** command to test the connectivity.

```
[R2]ping 10.0.23.3
  PING 10.0.23.3: 56   data bytes, press CTRL_C to break
    Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=20 ms
    Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=255 time=30 ms
    Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=255 time=30 ms
    Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=255 time=20 ms
```

```
    Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=255 time=30 ms

  --- 10.0.23.3 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/26/30 ms

[R2]
```

# 4 Verification

Omitted

# 5 Quiz

1. (Multiple-answer question) Which of the following statements are true about PPP? (ABDE)
   A. PPP supports the bundling of multiple physical links into a logical link to increase the bandwidth.
   B. PPP supports plaintext and ciphertext authentication.
   C. PPP cannot be deployed on Ethernet links because of its poor scalability.
   D. PPP supports asynchronous and synchronous links for the physical layer.
   E. PPP supports multiple network layer protocols, such as IPCP.
2. What phases does the PPP link establishment process include?
   Link layer negotiation, (optional) authentication negotiation, and network layer negotiation
3. Which protocol is used to negotiate IP parameters during the establishment of a PPP link?
   IPCP